SCS 3313: Applied Machine Learning

Practical: Feature Engineering

Feature Creation, Feature Transformation, and Feature Encoding

Instructions

- Complete the practical using Google Colab.
- Write your code in code cells and short explanations in text cells.
- Keep the original columns when creating or transforming features, unless a question specifically asks you to select a final set of columns.
- Run your cells and inspect the outputs.
- Model training is not required.

Part I

Create a new Google Colab notebook and run the following code.

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler, StandardScaler

data = {
    "Student_ID": [
        "S01", "S02", "S03", "S04",
        "S05", "S06", "S07", "S08"
    ],
    "Study_Hours": [4, 8, 6, 12, 10, 3, 14, 9],
    "Study_Days": [2, 4, 3, 4, 5, 1, 7, 3],
    "Sessions_Attended": [6, 8, 7, 10, 9, 5, 10, 8],
    "Total_Sessions": [10, 10, 10, 10, 10, 10, 10, 10],
    "Monthly_Allowance": [
        3000, 5000, 4000, 8000,
        6000, 2000, 25000, 7000
    ],
    "Study_Mode": [
        "Online", "Physical", "Hybrid", "Physical",
        "Online", "Hybrid", "Physical", "Online"
    ],
    "Confidence": [
        "Low", "Medium", "Medium", "High",
        "High", "Low", "High", "Medium"
    ],
    "Registration_Date": [
        "2026-01-05", "2026-01-10",
        "2026-02-03", "2026-02-14",
        "2026-03-02", "2026-03-07",
        "2026-04-06", "2026-04-12"
    ]
}

df = pd.DataFrame(data)
df
```

Question 1: Creating Numerical Features

1. Create a feature named `Average_Hours_Per_Day` using the following formula:

$$\text{Average Hours Per Day} = \frac{\text{Study Hours}}{\text{Study Days}}$$

2. Create a feature named `Attendance_Percentage` using the following formula:

$$\text{Attendance Percentage} = \frac{\text{Sessions Attended}}{\text{Total Sessions}} \times 100$$

3. Create an interaction feature named `Study_Attendance_Interaction`:

$$\text{Study Attendance Interaction} = \text{Study Hours} \times \frac{\text{Attendance Percentage}}{100}$$

4. Display `Student_ID` and the three new features.

5. Explain why average study hours per day could provide useful information beyond total study hours.

Question 2: Date and Aggregation Features

1. Convert `Registration_Date` to a pandas datetime column.

2. Extract the month number into a new column named `Registration_Month`.

3. Create a feature named `Weekend_Registration` containing:

   - `1` if registration occurred on Saturday or Sunday.
   - `0` otherwise.

4. Use `groupby()` to calculate the mean `Study_Hours` for each `Study_Mode`.

Question 3: Transforming Numerical Features

For this exercise, treat the entire dataset as a **training dataset**.

1. Use `MinMaxScaler` to transform `Study_Hours` into a new column named `Study_Hours_MinMax`.

2. Use `StandardScaler` to create a new column named `Study_Hours_Standardized`.

3. Apply `np.log10()` to `Monthly_Allowance` and save the result as `Log_Allowance`.

4. Create `Study_Hours_Squared` by squaring `Study_Hours`.

5. Display the original features beside their transformed versions.

6. Answer the following questions briefly:

   (a) What are the minimum and maximum values of `Study_Hours_MinMax`?

   (b) Does a negative standardized value mean that the original study hours were negative?

   (c) What happens to the relative size of the largest allowance after log transformation?

Question 4: Encoding Categorical Features

1. Identify whether `Confidence` and `Study_Mode` are **ordinal** or **nominal** features.
2. Encode `Confidence` into a new column named `Confidence_Encoded` using the following mapping:

Confidence	Encoded Value
Low	0
Medium	1
High	2

3. Use `pd.get_dummies()` to one-hot encode `Study_Mode`. Use `dtype=int` so that the new columns contain `0` and `1`.
4. Display the encoded results.
5. Explain why assigning `Online = 0`, `Physical = 1`, and `Hybrid = 2` could introduce a misleading relationship.

Question 5: Review

Answer each question in one or two sentences.

1. How does feature creation differ from feature transformation?
2. Why would `Student_ID` usually be excluded from model inputs?
3. When using separate training and test datasets, why should a scaler be fitted only on the training data?